

GUMI, Korea, Republic of

WMO: 471540

Lat: 36.1306N

Lon: 128.3206E

Elev: 165

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 18.9	(d) -6.0	(e) 4.0	(f) 26.1	(g) -2.2	(h) 4.9	(i) 26.7	(j) 13.7	(k) 27.8	(l) 12.2	(m) 29.4	(n) 4.7	(o) 290	(p) 0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.2	(c) 92.8	(d) 77.7	(e) 90.6	(f) 76.6	(g) 88.2	(h) 74.7	(i) 81.3	(j) 87.5	(k) 80.0	(l) 86.5	(m) 78.6	(n) 84.5	(o) 3.8	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.9	156.1	83.9	78.3	148.1	82.9	77.0	141.4	82.1	45.3	88.2	43.8	86.9	42.3	84.2	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.0	(b) 10.5	(c) 9.1	(e) 11.5	(f) 95.8	(g) 4.5	(h) 1.8	(i) 8.2	(j) 97.1	(k) 5.6	(l) 98.2	(m) 3.1	(n) 99.2	(o) -0.2	(p) 100.5		
			DB	WB	9.5	81.7	3.9	2.4	6.6	83.4	4.4	84.8	2.2	86.1	-0.7	87.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	31.2	36.5	45.7	56.0	66.8	73.9	79.2	79.6	70.5	59.3	47.1	34.5
	DBStd	17.71	6.08	6.52	6.35	5.90	4.76	3.73	4.22	4.71	4.60	5.41	7.05	6.17
	HDD50	1772	582	382	167	17	0	0	0	0	0	4	139	481
	HDD65	4435	1047	799	598	272	36	1	0	0	8	190	538	946
	CDD50	4252	0	3	34	198	519	718	905	917	614	293	51	0
	CDD65	1441	0	0	0	3	91	269	440	452	172	14	0	0
	CDH74	12769	0	0	2	111	963	2108	4180	4438	934	33	0	0
	CDH80	4393	0	0	0	11	236	622	1570	1791	163	0	0	0
(14) Wind	WSAvg	3.3	4.0	3.8	4.1	4.0	3.7	3.1	2.8	2.6	2.4	2.6	3.1	3.6
(15) Precipitation	PrecAvg	43.0	0.7	1.1	1.9	3.0	3.3	4.9	9.3	9.6	5.6	1.7	1.4	0.7
	PrecMax	68.7	2.3	3.5	4.4	6.5	8.1	13.8	25.2	23.6	17.3	5.4	5.2	1.9
	PrecMin	25.1	0.0	0.0	0.2	0.6	0.8	0.4	3.8	2.0	0.4	0.0	0.0	0.0
	PrecStd	10.0	0.5	0.9	1.1	1.6	1.7	3.0	4.8	5.2	4.3	1.5	1.2	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	62.5	72.2	81.2	89.8	91.0	94.7	96.2	87.9	77.3	70.0	54.1
		MCWB	40.5	54.7	55.1	56.5	64.8	68.3	79.8	79.1	74.5	66.2	59.7	48.6
	2%	DB	46.5	55.9	67.3	77.2	85.3	88.0	92.3	93.7	83.8	74.4	64.6	50.2
		MCWB	38.6	45.7	50.8	56.8	62.3	68.0	78.6	78.7	71.5	64.2	56.2	42.6
	5%	DB	43.6	51.4	63.2	73.6	82.4	86.0	90.2	91.6	81.6	71.8	61.5	47.3
		MCWB	36.7	41.7	48.8	54.7	61.6	68.2	77.3	78.0	70.4	61.5	54.5	41.5
	10%	DB	41.0	47.7	58.8	69.5	79.7	83.8	87.9	88.9	79.3	69.3	58.4	44.7
		MCWB	34.6	39.0	47.8	53.1	61.0	67.6	76.4	77.3	69.6	60.9	51.9	39.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	57.8	58.7	63.3	74.1	76.5	83.0	82.6	78.7	72.5	62.2	51.2
		MCDB	48.3	61.3	64.9	72.7	76.4	80.8	87.6	88.4	81.2	74.6	66.9	53.6
	2%	WB	40.4	47.8	55.3	60.2	67.6	74.0	81.7	81.3	76.1	68.9	58.6	45.6
		MCDB	44.3	51.1	60.3	68.6	76.3	79.6	87.0	88.3	80.7	71.2	62.9	48.1
	5%	WB	37.9	43.1	52.3	58.2	65.5	72.2	80.5	80.1	74.2	65.3	55.9	42.2
		MCDB	42.0	49.6	62.5	67.1	75.3	79.0	86.5	87.5	78.5	68.2	59.6	46.4
	10%	WB	35.5	40.1	48.1	55.7	63.7	70.7	79.1	78.9	72.1	62.1	53.7	39.6
		MCDB	40.1	46.2	56.7	65.4	74.0	78.4	84.4	85.6	76.5	67.2	57.1	44.2
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.2	19.9	21.4	22.0	17.2	13.4	14.2	16.0	18.7	16.9	15.0
		MCDBR	18.7	22.7	27.6	30.0	27.7	21.5	17.2	18.4	18.7	21.4	20.3	18.5
	5% WB	MCWBR	12.3	14.0	14.0	12.4	9.3	5.9	5.2	5.5	7.6	9.9	11.4	12.5
		MCDWR	16.9	20.1	22.8	21.6	20.3	14.2	14.4	15.8	13.5	12.9	16.4	16.2
(40) Clear-Sky Solar Irradiance	taub	taub	0.405	0.454	0.536	0.570	0.580	0.637	0.593	0.549	0.480	0.455	0.430	0.399
		taud	2.039	1.950	1.786	1.761	1.780	1.723	1.892	2.002	2.141	2.083	2.065	2.075
	Ebn at Noon	Ebn at Noon	238	240	232	233	233	220	228	236	246	238	227	230
		Edh at Noon	41	50	65	70	70	74	62	55	45	44	40	37
(44) All-Sky Solar Radiation	RadAvg	RadAvg	846	1073	1384	1582	1775	1612	1386	1396	1245	1144	845	757
	RadStd	RadStd	53	100	132	121	163	128	188	185	141	99	102	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	N/A

Nomenclature: See separate page